

Kannada Speakers Speech Emotion Recognition Using Deep Learning Technique: Anger, Fear, Sad and Neutral state

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ABSTRACT

In this paper, we have experimenting to identifying the Kannada emotional speech signals recognition using deep learning technique. Speech Emotion Recognition (SER) is a hot research topic in the field of HCI (Human Computer Interaction). To experiment we have created our own Kannada Emotional Speech corpus recording in different android mobile phone, with sampling rate 48 kbps and mono channel and speakers are from the Karnataka and Tamil Nadu boarder region, whose accents is a mixer of Kannada, Tamil and Telugu languages. The emotions are anger, fear, sad and neutral is considered. Totally we have 440 Kannada emotional speech signals. To use this speech signal data in our experiment, we have first down sampled all speech signals into 16 kbps, mono channel and stored as wav files. Then we pass the each sample signals into the pre-emphasize phase, then framing and windowing is done, then we extracted features from each voice frame signal like MFCC, Chroma and 'Mel' and then passed to CNN (Convolution Neural Network) to train and testing the voice signals. The CNN is capable to handle small amount of data for training, and with smaller number of parameters. The average Kannada emotion recognition accuracy rate of 93.77% is achieved. All computations are done using Python programming language and used Python libraries like Librosa, Keras, pyaudio, soundfile, sklearn.

Keywords: Accent, Emotion Recognition, MFCC, CNN, HCI, Librosa

Introduction

Language is one of the most important methods for communication and speech is one of its main mediums. Designing the human to machine interface, the speech signal is transformed into analogue and digital waveform which can be understood by the

machine. Speech technologies are broadly used and seen to have unlimited uses. In many of the human-machine interface applications, emotion recognition from the speech signal is considered to be the research topic for many years. For the identification of the emotions from the speech signal, many systems have

been developed until now. In this paper, speech emotion recognition based on the previous technologies which use different models and methods for the emotion recognition is reviewed and a new approach is suggested. They are used to differentiate emotions such as anger, sad, fear and neutral state when accents are mixtures of Kannada, Tamil and Telugu languages. Accent is a way of speaking usual of a particular group of people and especially of the natives or residents of a region. In Kannada language, the 'accent' means 'Ottu' and 'Uchchaaraṇe', referring to pronunciation, stress, or a regional way of speaking, which shows a distinct sound pattern in their words pronunciation. It also shows the importance given to 'Ottu akashara' to pronunciation of words by stressing or emphasising on a syllable of the Kannada languages. Hence our experiment becomes important on recognition of the Kannada emotions on different accents speakers.

Emotion recognition in spoken dialogues has been gaining increasing interest all through current years. Speech Emotion Recognition (SER) is a hot research topic in the field of Human Computer Interaction (HCI). It has a potentially wide application, such as the interface with robots, banking, call centers, car board systems, computer games, surveillance systems, psychological treatments, robotics, online marketing etc. For classroom orchestration or E-learning, information about the emotional state of students can provide focus on enhancement of teaching quality. For example teacher can use SER to decide what subjects can be taught and must be able to develop strategies for managing emotions within the learning environment.

Librosa is a python Library module used for investigating the music and sound which gives us the referral devices we want to design and develop. It absorbs the sound documents as the input for the examination and modifies them with the help of Fourier transform techniques. It also plots the outcomes of the model.

LITERATURE SURVEY

Bennilo Fernandes (2021) et al., have evaluated using RNN with hierarchal of LSTM and BiLSTM with dropout layers and compared their performances on interchanging the layers for a reduced Tamil emotional speech data set. Also they evaluated different architectures like; DHLL, DHLB, DHBL and DHBB with dropout layers and the evaluation were accuracy, loss, training time and evaluation time. The results showed that the DHLB performs better than other models. Accuracy rate of 84% is achieved with minimum loss in each seven basic emotions and time taken for training and evaluation is also less than the other models. Thus, they recommended for the reduced Tamil emotional speech data set is to use DHLB, since it has returned good efficiency of recognition values within an acceptable running time. Javier G. Razuri (2015) et al., experimented on the human's nonverbal cues such as pitch, loudness, spectrum, speech rate which are the efficient carriers of emotions. Linear predictive coding (LPC) is used to estimate the basic parameters into a speech signal, such as the vocal tract transfer function and the formant frequencies. It has good intuitive interpretation both in time domain and in frequency domain. The cepstral representation (Linear Prediction Cepstral Coefficients (LPCC)) of its coefficients is more used because of its higher retrieval efficiency. Chunyi Wang, presented acoustic features typically utilized in speech emotion recognition such as time-domain features, frequency-domain features, statistical features, deep features and hybrid features. In general, feature extraction depends on the frame involved. A speech signal with a certain duration is divided into frames with an interval ranging from 10 to 30ms, while features are then extracted from every frame. Time-domain feature is the direct process of the time domain waveform. This is the most prevalent feature extraction method since it is relatively easy to think of. It can extract features such as short-time zero crossing rate, short-time energy and pitch frequency. Frequency domain

feature involves (short-time) Fourier transform wavelet transform and others. Initially, the time-domain signal is transformed into the frequency-domain, from which the feature is extracted. Such features are highly associated with the human perception of speech. Hence, they have apparent acoustic characteristics. These features are usually comprised of formant frequency, linear prediction cepstral coefficient (LPCC), and Mel frequency cepstral coefficients (MFCC). They can rightly signify the channel feature, indicating noise resistance and good recognition performance. Mohanty et al., (2022), fills as the establishment for advancements, for example, voice acknowledgment, voice control, order capacities, and a lot more presented by tech goliaths like Google and Microsoft as Alexa, Cortana, and Samsung's Bixby, as well as other man-made brainpower applications. Harshith B U et al., (2019) used the happy sentence to the system; they found more error comparably to that of angry and sad sentence. The fact for the reduced accuracy in happy sentence is that in angry sentence pitch will be high and for the sad sentence pitch will be low but in the happy sentence the pitch value will be in between the other two emotions. However, the main feature we considered is pitch and its parameter the accuracy will be dependent on pitch. Syeda Tamanna Alam Monisha et al., (2022), in their research papers shown that the world being digitized day by day, speech emotion recognition has found increasing applications in our daily lives. Geethashree and D. Ravi, (2018), used the Self-created database containing 2800 emotional recordings Kannada Bayesian and HMM model with MFCC feature Happy, excited, angry, and sad found that the Average emotion error rate of 25.5% for Bayesian and 0.2% for HMM approach. Rahul. B. Lanjewar et al., (2013) used the features like the pitch related statistics, Formants and Mel Frequency Cepstral Coefficients (MFCC) are also frequently found to contribute as feature vectors. The parametric representations of Spectral measures other than MFCCs are also common in research including the

features of Wavelets, Teager energy operator (TEO) based features, Log Frequency Power Coefficients (LFPC) and Linear Prediction Cepstral Coefficients (LPCC). These extracted features are stored by training the database for the classification stage. Classifiers to detect emotions, feature vectors belong to the global statistics SVM (Support Vector Machine), Neural Networks, Decision trees are employed and for the vectors of short-term features HMM (Hidden Markov Model) is used for its dynamic performance. More diverse results have obtained in terms of robustness when mathematical Teager Energy Operator (TEO) were employed on speech to calculate their energy that reflects the stress in the airflow structure of speech for both text dependent and text independent models. Xu Dong An and Zhou Ruan (2021), used the traditional phonological features by dividing into acoustic features, rhythmic features and phonological features. Acoustic features are mainly divided into vowels and consonants in speech; rhythmic features are divided into energy, speech rate and resonance peaks; and phonological features represent the clarity of the speech signal. Due to the complexity of speech emotion, many emotions are difficult to be recognized effectively by rhyme and sound quality alone, resulting in poor differentiation of a single original feature, which requires the confusion of different speech features for speech recognition. Moreover, the speech signal is a non-smooth random process with strong time variability, so to increase the practicality of feature parameters and reduce the complexity of feature extraction, MFCC is selected as the speech emotion feature in their paper. Shubham Singh Chaudhary and Sachin Garg (2021), discussed on the developments of AI giving rise to several technologies involved in Human-Computer Interaction. Aiming to develop and upgrade HCI methods is of foremost importance because HCI is the frontend of AI which millions of users experience. Some of the present HCI methods require communication through touch, movement, hand gestures, voice and facial gestures. Amid the

different methods, the voice-based intelligent devices are acquiring popularity in a board range of applications. In a voice-based system, a computer agent is required to completely apprehend the human's speech form in order to precisely pick up the commands given to it. This field of study is called as Speech Processing and involves of three components like Speaker Identification, Speech Recognition and Speech Emotion Detection. Speech Emotion Detection is challenging to execute amidst the other components due to its complexity. Moreover, the definition of an intelligent computer system needs the system to copy human behavior. A salient nature unique to humans is the capability to alter discussions based on the emotional state of the speaker and the listener. Speech emotion detection can be made as a classification issue solved using many machine learning algorithms. They created a Multi Perceptron model, LSTM (Long short term memory) model and CNN models. The MLP and LSTM were not suitable as it gave us low accuracy in their experiment.

SPEECH CORPUS

In this experiment, we have designed our own Kannada emotional speech signals database. We have recorded the Kannada sentence as shown in table-1 using the different android mobile recorder. Here emotions used are Anger, Fear, Sad and Neutral (Normal) voice. Android mobile recorded signals are at the rate of 48Kbps and mono channel. Here the speakers are the people living near boarder region of Karnataka and Tamil Nadu area that is Chamarajanagara district forest region boarded to two states. Table 2 shows that number of speakers used to record the Kannada sentences in different emotions. Here each sentence is uttered by 5 times by each speaker, making 5 emotional signals of each type. Totally we have used 440 Kannada emotional speech signals recorded in different android mobile phone has different frequency. Hence to make uniform frequency, each signal is resampled to 16 kbps, mono

channel and stored as wav files before using the emotional speech signals in our experiment.

Table 1: Kannada Sentence used to design the Emotion Speech Signal. Its continuous Kannada speech signals.

Sl. No	Sentence in Kannada Script	Sentence in English Transcript words
1.	ಇದು ಕಲಿಯುಗ ಸಕ್ಕರೆ ಇದ್ರೇನೆ ನೊಣ, ಹಣ ಇದ್ರೇನೆ ಜನಾ	Idu kaliyuga sakkare idrēne noṇa, haṇa idrēne janaa
2.	ನಾನು ಶಾಲೆಗೆ ಹೇಗುತ್ತೇನೆ ಆದರೆ ಅವನು ಮನೆಗೆ ಹೇಗುತ್ತಾನೆ	Naanu saalege hoogutteene aadare avanu manege hooguttaane

In this experiment, we have used Intel CORE i5 7th generation laptop having processing capacity of 2.5GHz and 3 GB RAM. Here all our speakers are peoples living in the Karnataka and Tamilnadu boarder forest region, its located in chamarajanagara district, this region people basically speak Kannada language but they have learnt to speaks Tamil and Telugu languages from childhood, hence their accents is mixed with this 3 languages from their birth. Hence we have selected the speakers who are educated at least 10 + 2 level or more educated to read the Kannada sentence shown in our table 1 and produce the emotional speech.

Table 2: Shows that number of speakers used to record the Kannada sentences in different emotions. Here each sentence is uttered by 5 times by each speaker, making 5 emotional signals of each type by each speaker. Totally we are having 440 signals or instances. Among 440 Kannada emotional speech signals used for 352 speech signals for training and testing used 88 samples speech signals.

Gender	Anger	Fear	Sad	Neutral
Male Speakers	14 speakers	14 speakers	14 speakers	14 speakers
Female Speakers	10 speakers	10 speakers	10 speakers	10 speakers

METHODOLOGY

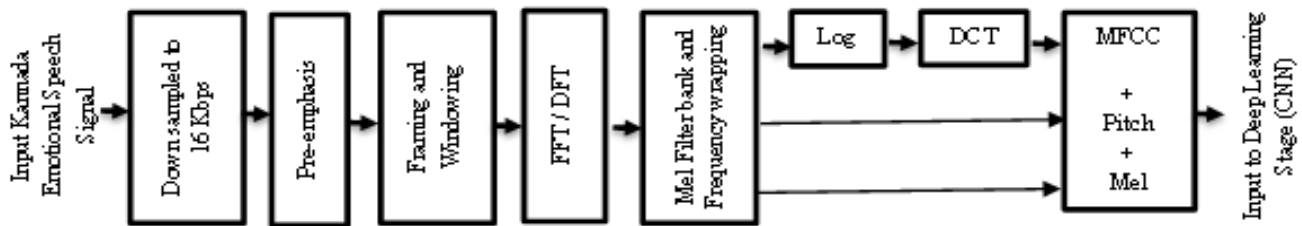


Figure 1: Show the Proposed Model for down sampling our Kannada emotional speech signal, pre emphasizing phase, framing and Windowing, and then from each voiced signal frame extracting the features like MFCC, Pitch and Mel values and combined to form row then passed to our deep learning model (CNN) for training the data using different epochs.

In this experiment, each Kannada emotional speech signal sample is stored as wav file, and which is down sampled at 16000 Hz. Then signal is passed with pre-emphasis phase, here we observe the spectrum of voiced speech segment that is high frequencies usually have smaller amplitude of high frequency bands and decrease the amplitude of the lower bands making the spectrum flatter. Then, frame blocking which refers to the process of segmenting the speech signal after pre-emphasis into number of small overlapping frames. We have taken frame size of 20 msec, with an overlap of 6.5 msec [19]. Then each frame is passed through the Hamming Window, which extracts the stationary part of speech [15][17][19]. Next is features extraction stage, here we have extracting MFCC (Mel Frequency Cepstral Coefficient) represents the short-term power spectrum of a sound, Chroma which concerns to the 12 different pitch classes and 'Mel' which is Mel Spectrogram Frequency. We extracted 26 features values from each voice signal frame [15][16][17][19],

which is passed to the CNN model for training the data. Her parameter setting is required before passing to CNN, those modifications are

Data tensor matrix in our experiment: After pre-processing, we obtain N frames of each voice sample and 13 MFCCs, 12 pitches and one 'Mel' value for each frame. Here each following frame is dependent on previous frame, which is termed as time dependency or temporal nature. For each voice samples, we have 2-D data matrix where rows represent contents of each frame (13 MFCC + 12 Pitches + 1 Mel = 26 values) and columns represent number of frames that are dependent sequentially. So matrix size will be 26 * N for each instance of voice sample.

Hyper Parameters: Then we need tuning of hyper parameters to achieve desired accuracy. Here we have directly used Keras library which gives flexibility of tuning of hyper parameters. Firstly, batch size used is 32 which is default batch size of Keras then applied Adam optimizer. Then Dropout regularization, which deals with over fitting in neural networks, here we

used 0.1 as dropped dimensions. Then used 3 stacked layers.

Number of Recurrent Cells: In our experiment we have 'N' frames (Where 'N' varies for each instance of voice signal, because we have taken Kannada emotional sentences uttering has continuous speech signal) to deal with temporal nature of the application, we have used 'N' recurrent cells.

Epochs: The numbers of epochs are defined as the number of times we are training our model. We are using Early Stopping which is used in LSTM to remove over fitting.

Using those parameters for CNN each Kannada emotional signals voice part is trained and stored as features values of each type of emotional signals separately. The CNN itself used here as classifier, using this each type of emotional data is classified and

stored using labels. In our experiment CNN is used for testing the Kannada emotional speech signal.

EXPERIMENTAL RESULTS

In this paper we have experimented only for Kannada emotional speech like anger, fear, sad and neutral state speaking. Here CNN model is used for training and testing purpose with different feature sets of data and with different epochs. The table 3 shows the accuracy rate of recognition and identification of Kannada emotion speech of having CNN model for testing and training, but with different features set values and epoch is kept 400 as constant. The table 4 shows the proposed mode with different epochs for Kannada emotion speech signals like anger, sad, fear and neutral. The average Kannada emotion recognition accuracy rate of 93.77% is achieved in this experiment.

Table 3: Showing the extracted features to pass into the CNN model for training the data and stored with labelled. Here the accuracy rate of recognition and identification of Kannada emotion speech in this table according to emotions used here is anger, fear, sad and neutral state and compared the accuracy rate with different features values. Here epochs of 400 is used.

Features values passing to CNN	Anger	Fear	Sad	Neutral
MFCC values	79.12%	78.34%	78.11%	79.89%
Pitch values	78.67%	78.39%	78.12%	78.99%
MFCC + Mel values	79.07%	79.23%	79.43%	79.67%
Pitch + Mel values	78.37%	78.2%	78.92%	79.39%
MFCC + Pitch values	89.21%	86.89%	87.02%	92.58%
Proposed method values	95.39%	91.36%	91.38%	96.96%

Table 4: Showing the experimental results on proposed mode with different epochs for Kannada emotions speech signals like anger, sad, fear and neutral.

No. of Epochs	Anger	Fear	Sad	Neutral
100 Epochs	92.01%	90.98%	90.9%	93.78%
200 Epochs	94.23%	91.04%	91.23%	95.34%
300 Epochs	95.12%	91.34%	91.35%	96.45%
400 Epochs	95.39%	91.36%	91.38%	96.96%
500 Epochs	95.38%	91.36%	91.37%	97.16%

DISCUSSION AND CONCLUSION

In our experiment we have used only 440 emotional Kannada speech signal of only one regional accents and trained and tested, hence we have received only 93.77% of accuracy when the epochs is 400. If consider the all Kannada accents and trained with the more number of samples signals than may expect the more accuracy with less epochs and system can be used emotional detection in real world applications. In this experiment we have used the feature like MFCC, Pitch and Mel. The MFCC are claimed to be robust of all the features for any speech tasks. The consonant-vowel transition region also yields emotion recognition but when formants features combined with the basic spectral features have always improvement in the system performance. We absorbed that CNN technique is better technique in training and testing for recognition of any Kannada emotional speech signals. But only thing is that we need to apply all types of Kannada accents for training to build Kannada emotion recognition for any Kannada accents.

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References

- [1]. Bennilo Fernandes et al., (2021), "Speech Emotion Recognition Using Deep Learning LSTM for Tamil Language", *Pertanika J. Sci. & Technol.* 29 (3): 1915 - 1936 (2021), e-ISSN: 2231-8526, DOI: <https://doi.org/10.47836/pjst.29.3.33>
- [2]. Leila Kerkeni et al., (2018), "Speech Emotion Recognition: Methods and Cases Study", In *Proceedings of the 10th International Conference on Agents and Artificial Intelligence (ICAART 2018)* - Volume 2, pages 175-182 ISBN: 978-989-758-275-2.
- [3]. Javier G. Razuri et al., (2015), "Speech emotion recognition in emotional feedback for Human-Robot Interaction", (*IJARAI*) *International Journal of Advanced Research in Artificial Intelligence*, Vol. 4, No.2, 2015.
- [4]. Chunyi Wang et al., (2022), "Speech Emotion Recognition Based on Multi-feature and Multilingual Fusion", *Multimedia Tools and Applications*, 2022, Springer, DOI:10.1007/s11042-021-10553-4
- [5]. Mohanty et al., (2022), "Speech Emotion Recognition System using Librosa for Better Customer Experience," *Graduate Research in Engineering and Technology (GRET): Vol. 1: Issue 6, Article 7.* DOI: 10.47893/GRET.2022.1114.
- [6]. Harshith B U et al., (2019), "Speaker Dependent Emotion Recognition from Speech for Kannada language", *International Journal of Engineering Research & Technology (IJERT)* ISSN: 2278-0181 Published by, www.ijert.org NCRACES - 2019 Conference Proceedings, Special Issue - 2019, Volume 7, Issue 10, Pg 1-4.
- [7]. Syeda Tamanna Alam Monisha et al., (2022), "A Review of the Advancement in Speech Emotion Recognition for Indo-Aryan and Dravidian Languages", *Hindawi Advances in Human-Computer Interaction* Volume 2022, Article ID 9602429, 11 pages. <https://doi.org/10.1155/2022/9602429>
- [8]. Geethashree and D. Ravi, (2018), "Kannada emotional speech database: design, development and evaluation," in *Proceedings of the International Conference on Cognition and Recognition*, pp. 135-143, Springer, 2018.
- [9]. Ashish B. Ingale et al., (2012), "Speech Emotion Recognition", *International Journal of Soft Computing and Engineering (IJSCE)* ISSN: 2231-2307, Volume-2 Issue-1, March 2012 pg 235-238.

- [10]. Xu Dong An and Zhou Ruan (2021), "Speech Emotion Recognition algorithm based on deep learning algorithm fusion of temporal and spatial features", Journal of Physics: Conference Series 1861 (2021) 012064, IOP Publishing doi:10.1088/1742-6596/1861/1/012064.
- [11]. Xingyu Cai et al., (2021), "Speech Emotion Recognition with Multi-task Learning", INTERSPEECH 2021 30 August – 3 September, 2021, Brno, Czechia, pg 4508-4512.
- [12]. Rahul. B. Lanjewar et al., (2013), "Speech Emotion Recognition: A Review", International Journal of Innovative Technology and Exploring Engineering (IJITEE) ISSN: 2278-3075 (Online), Volume-2 Issue-4, March 2013, Pg. 68-71.
- [13]. Siddhant S. Patil et al., (2022), "Speech Emotion Recognition System Using Recurrent Neural Network in Deep Learning", International Journal for Research in Applied Science & Engineering Technology (IJRASET), Volume 10 Issue III Mar 2022, pg. 2332-2338, ISSN: 2321-9653.
<https://doi.org/10.22214/ijraset.2022.41112>.
- [14]. Shubham Singh Chaudhary and Sachin Garg, (2021), "Speech Emotion Recognition", International Research Journal of Modernization in Engineering Technology and Science, Volume 03, Issue: 12, December-2021, pg. 218-221, e-ISSN: 2582-5208.
- [15]. Hemakumar G. et al., (2016), "Large Vocabulary in Continuous Speech Recognition Using HMM and Normal Fit", International Journal of Computer Trends and Technology (IJCTT) – Volume 42 Number 2 – December 2016 Pg. 102-107, ISSN: 2231-2803.
- [16]. Hemakumar G. et al., (2014) "Speakers' Accent And Isolated Kannada Words Recognition", American Journal of Computer Science & Information Technology, Publishing by PUBICON International Publication, Volume 2, Issue 2, Nov-Dec 2014, pg. no. 071-077, ISSN: 2349-3917.
- [17]. Hemakumar G et al., (2025), "Large Language Model For Kannada Speech Recognition", International Journal of Engineering Development and Research, July 2025, Volume 13, Issue 3, Pg 603 – 609, ISSN: 2321-9939.
- [18]. M. Gopal, (2025), "Deep Learning Core Concepts, Methods and Application", publication by Pearson India Education Services Pvt. Ltd, fourth impression 2025, ISBN 978-93-560-6197-2.
- [19]. Hemakumar G. et al., (2014) "Automatic Segmentation of Kannada Speech Signal into Syllables and Sub-words: Noised and Noiseless Signals", published by International Journal of Scientific & Engineering Research, Volume 5, Issue 1, January-2014, Page No 1707-1711, ISSN 2229-5518.